

JAINDU SAMARANAYAKE

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EDUCATION

Bachelor of Science in Electronic and Telecommunication Engineering University of Moratuwa, Sri Lanka	March 2024 - Present CGPA: 3.6/4.00
Upper Secondary Education - Physical Stream Central College, Kuliyaipitiya	June 2020 - December 2022 [G.C.E Advanced Level Examination - 3A's]

TECHNICAL SKILLS & INTERESTS

Programming & Web	Python, Java (Spring Boot), C++
Hardware & Embedded	ESP32, Raspberry Pi, STM32L0, Arduino Mega, DE0-Nano (FPGA), I2C
Engineering Software	SolidWorks, Altium, Intel FPGA (Quartus), MATLAB/Simulink
Domains of Interest	Machine Learning, Automation, Robotics, Deep Learning, LLMs, IoT

PROJECTS

Dengue Patient Monitoring System (Team NEXORA) | SPARK Challenge November 2025 - Present

- Developing an automated medical monitoring system utilizing a centralized hub and sensor networks.
- Optimizing current diagnostic workflows by integrating Machine Learning for data analysis to reduce human interaction and improve patient outcomes.

Autonomous Robotics (SLRC 2026 & Robot Design) | Academic & Competition July 2025 - Present

- Handled the virtual simulation and automation components for the Sri Lanka Robotics Challenge (SLRC) 2026 robot.

Smart Water Management System | Academic Group Project

- Developed an automated smart water meter integrated with a web application for real-time data monitoring and resource management.

Laboratory Database Management System | Academic Project

- Developed a full-stack laboratory management platform utilizing Java Spring Boot for the backend API and developed the frontend admin dashboard.

Enterprise Network Design | Academic Assignment March 2026

- Designed the backbone network architecture for the University of Moratuwa and the internal network for the ENTC department building.

Hardware Product Development (Sound Regulation IoT) | Curriculum Projects February 2025 - November 2025

- Designed an IoT device and an analog circuit to divide sound inputs into bands and regulate microphone feed, including custom PCB layout and enclosure design.

POSITIONS OF RESPONSIBILITY & EXTRACURRICULAR

- **Committee Member, IEEE Robotics and Automation Society (RAS)** November 2025 - Present
Organized MAZEX 1.0, a prominent Micromouse competition and associated technical workshop series.
- **Organizing Committee Member, ENTC Career Fair '26** January 2026
- **Organizing Committee Member, Sri Lanka Robotics Challenge** April 2025
- **Mentor, Soyuru Sathkara Seminar Series** January 2025